

DBMS Practical Exam - Cheat Sheet

1. SQL Queries - Example (Student & Employee)

-- Student Table Query

```
SELECT * FROM Student WHERE age > 20;
```

-- Employee Table Query

```
SELECT name, salary FROM Employee WHERE department = 'IT';
```

2. Join Queries - Doctor & Hospital

```
SELECT Doctor.name AS Doctor_Name, Hospital.name AS Hospital_Name  
FROM Doctor  
JOIN Hospital ON Doctor.hospital_id = Hospital.hospital_id;
```

3. Views

-- Creating a view for high salary employees

```
CREATE VIEW HighSalaryEmp AS  
SELECT * FROM Employee WHERE salary > 50000;
```

-- Querying the view

```
SELECT * FROM HighSalaryEmp;
```

4. PL/SQL Blocks

-- Basic block

```
DECLARE  
    emp_id INT := 2;  
    emp_name VARCHAR(50);  
BEGIN  
    SELECT first_name INTO emp_name FROM Employee WHERE employee_id = emp_id;  
    DBMS_OUTPUT.PUT_LINE('Employee name is ' || emp_name);  
END;
```

-- Conditional block

```
DECLARE  
    sal NUMBER := 40000;  
BEGIN  
    IF sal > 50000 THEN  
        DBMS_OUTPUT.PUT_LINE('High Salary');  
    ELSE  
        DBMS_OUTPUT.PUT_LINE('Average Salary');  
    END IF;  
END;
```

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5. Triggers

```
-- Before Insert Trigger
CREATE OR REPLACE TRIGGER trg_before_insert
BEFORE INSERT ON Student
FOR EACH ROW
BEGIN
    DBMS_OUTPUT.PUT_LINE('Inserting student: ' || :NEW.first_name);
END;

-- After Update Trigger
CREATE OR REPLACE TRIGGER trg_after_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    DBMS_OUTPUT.PUT_LINE('Updated employee ID: ' || :OLD.employee_id);
END;
```

6. Viva Questions

Q: What is a View?

A: A virtual table created by a query.

Q: Types of Joins?

A: INNER, LEFT, RIGHT, FULL JOIN.

Q: What is a Trigger?

A: A block that auto-executes on DML actions.

Q: What is PL/SQL?

A: Oracle's procedural SQL extension.

Q: DELETE vs TRUNCATE?

A: DELETE is slow and rollback-able; TRUNCATE is fast but permanent.

Q: What is normalization?

A: Organizing data to reduce redundancy.

Q: Aggregate functions?

A: COUNT(), SUM(), AVG(), MAX(), MIN().